

# A Superfast Direct Solver for 2D Type-II Inverse Nonuniform Discrete Fourier Transform Based on Hierarchically Semiseparable Matrices

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Joint work with Prof. Yingzhou Li

- 1 Introduction
- 2 Review of the 1D Method
- 3 Low-Rank Property of the NUDFT Matrix
- 4 Approximate Direct Inversion of the 2D Type-II NUDFT Matrix
- 5 Numerical Examples

Two-dimensional **nonuniform discrete Fourier transform (NUDFT)**:

$$f_j = \sum_{k^{[x]}=0}^{n^{[x]}-1} \sum_{k^{[y]}=0}^{n^{[y]}-1} c_{k^{[x]}, k^{[y]}} e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)}, \quad 0 \leq j \leq M-1.$$

- Sample points:  $(x_j, y_j) \in [0, 1)^2$ ;
- Frequencies:  $(k^{[x]}, k^{[y]}) \in \{0, 1, \dots, n^{[x]} - 1\} \times \{0, 1, \dots, n^{[y]} - 1\}$ ;
- Coefficients:  $c_{k^{[x]}, k^{[y]}} \in \mathbb{C}$ ;
- Target values:  $f_j \in \mathbb{C}$ ;
- Number of sample points  $M$ , number of frequencies  $N = n^{[x]}n^{[y]}$ .

NUDFT matrix:

$$\mathbf{A} = \left( e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)} \right)_{j, (k^{[x]}, k^{[y]})} \in \mathbb{C}^{M \times N}.$$

# Forward and Inverse Problems

**Forward problem:**  $\mathbf{f} = \mathbf{A}\mathbf{c}$ .

- NUFFT [Barnett, Magland, and af Klinteberg 2019]  
Complexity:  $\mathcal{O}(M + N + N \log N)$ .

**Inverse problem:**  $\min_{\mathbf{c} \in \mathbb{C}^N} \|\mathbf{A}\mathbf{c} - \mathbf{f}\|_2, \mathbf{c} = \mathbf{A}^\dagger \mathbf{f}$ .

- Iterative methods: LSQR + NUFFT;
- Direct methods: [Kircheis and Potts 2023].

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# Main Idea

1D NUDFT matrix:

$$\mathbf{A} = \left( e^{-2\pi i k x_j} \right)_{j,k} \in \mathbb{C}^{M \times N} \quad \text{with} \quad x_j \in [0, 1).$$

1D DFT matrix:

$$\mathbf{F} = \left( \mathbf{F}(k, \ell) = e^{-2\pi i k \ell / N} \right)_{k,\ell} \in \mathbb{C}^{N \times N}.$$

The matrix  $\mathbf{G} = \mathbf{A}\mathbf{F}^{-1}$  can be expressed as a **kernel matrix**:

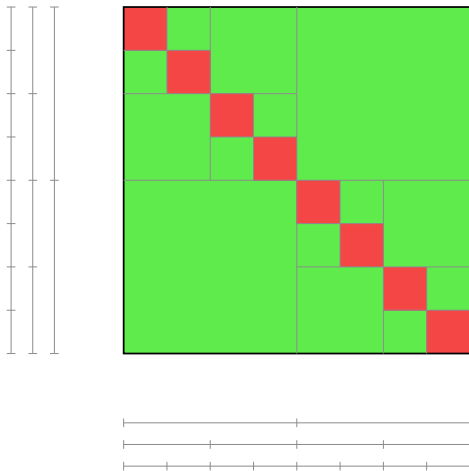
$$\mathbf{G}(j, k) = \frac{1}{N} \frac{(\gamma_j / \xi_k)^N - 1}{(\gamma_j / \xi_k) - 1} =: \Psi(\gamma_j, \xi_k).$$

- $\gamma_j = e^{-2\pi i x_j} \in \mathbb{S} := \{z \in \mathbb{C} : |z| = 1\};$
- $\xi_k = e^{-2\pi i k / N} \in \mathbb{S}.$

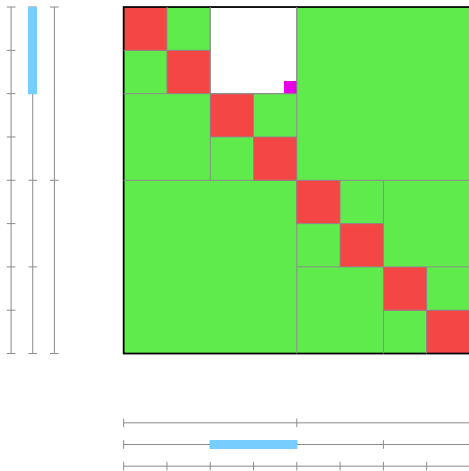
$\rightsquigarrow$  Approximate  $\mathbf{G}$  by a **hierarchically semiseparable (HSS) matrix**.

$\rightsquigarrow$  Use an HSS least-squares solver based on a URV factorization [Xi et al. 2014].

# Illustration of HSS Matrices

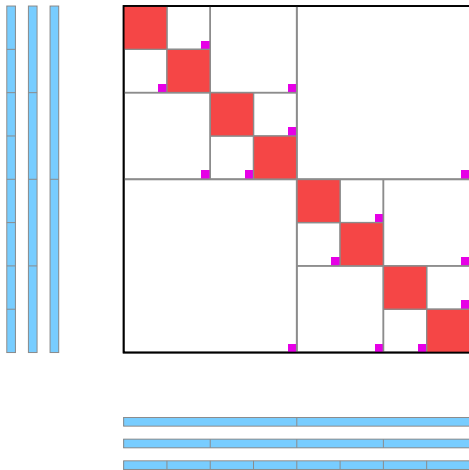


# Illustration of HSS Matrices

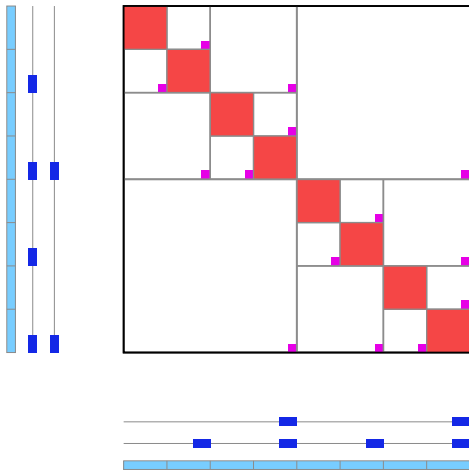


$$\begin{bmatrix} \text{Green Block} \end{bmatrix} = \begin{bmatrix} \text{Blue Bar} \end{bmatrix} \begin{bmatrix} \text{Magenta Square} \end{bmatrix} \begin{bmatrix} \text{Blue Bar} \end{bmatrix}$$

# Illustration of HSS Matrices



# Illustration of HSS Matrices



$$\begin{bmatrix} \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \end{bmatrix} = \begin{bmatrix} \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \end{bmatrix} \begin{bmatrix} \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \\ \text{blue bar} \end{bmatrix}$$

# Low-Rank Property of the Transformed NUDFT Matrix

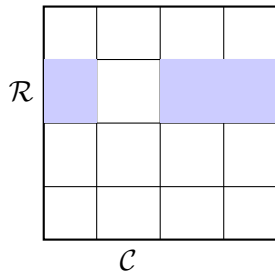
## Theorem (Numerical rank of HSS blocks)

Let  $[a, b) \subset [0, 1)$  be an interval, and define

$$\mathcal{R} = \{j : x_j \in [a, b)\} \subset \mathcal{J}, \quad \mathcal{C} = \{k : k/N \in [a, b)\} \subset \mathcal{K}.$$

Then, for  $0 < \varepsilon < 1$ , the HSS row block  $\mathbf{G}(\mathcal{R}, \mathcal{C}^c)$  has  $\varepsilon$ -rank at most  $\mathcal{O}(\log(1/\varepsilon) \log |\mathcal{C}|)$ .

*Wilber, Epperly, and Barnett 2025*



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# Face-Splitting of the NUDFT Matrix

NUDFT matrices in the  $x$ - and  $y$ -directions:

$$\mathbf{A}^{[x]} = \left( e^{-2\pi i k^{[x]} x_j} \right)_{j,k^{[x]}} \in \mathbb{C}^{M \times n^{[x]}} \quad \text{and} \quad \mathbf{A}^{[y]} = \left( e^{-2\pi i k^{[y]} y_j} \right)_{j,k^{[y]}} \in \mathbb{C}^{M \times n^{[y]}}.$$

Then

$$\mathbf{A} = \mathbf{A}^{[x]} \bullet \mathbf{A}^{[y]} = \begin{bmatrix} \mathbf{A}^{[x]}(0, :) \otimes \mathbf{A}^{[y]}(0, :) \\ \mathbf{A}^{[x]}(1, :) \otimes \mathbf{A}^{[y]}(1, :) \\ \vdots \\ \mathbf{A}^{[x]}(M-1, :) \otimes \mathbf{A}^{[y]}(M-1, :) \end{bmatrix} \in \mathbb{C}^{M \times N}.$$

# Transformed NUDFT Matrix

DFT matrices in the  $x$ - and  $y$ -directions:

$$\mathbf{F}^{[x]} \in \mathbb{C}^{n^{[x]} \times n^{[x]}} \quad \text{and} \quad \mathbf{F}^{[y]} \in \mathbb{C}^{n^{[y]} \times n^{[y]}}.$$

Let  $\mathbf{F} = \mathbf{F}^{[x]} \otimes \mathbf{F}^{[y]} \in \mathbb{C}^{N \times N}$  be the 2D DFT matrix. Then

$$\mathbf{G} = \mathbf{A}\mathbf{F}^{-1} = (\mathbf{A}^{[x]} \mathbf{F}^{[x],-1}) \bullet (\mathbf{A}^{[y]} \mathbf{F}^{[y],-1}) = \mathbf{G}^{[x]} \bullet \mathbf{G}^{[y]},$$

whose entries can be expressed as

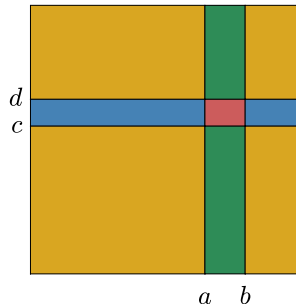
$$\mathbf{G}(j, (k^{[x]}, k^{[y]})) = \Psi^{[x]}(\gamma_j^{[x]}, \xi_{k^{[x]}}^{[x]}) \Psi^{[y]}(\gamma_j^{[y]}, \xi_{k^{[y]}}^{[y]}) = \Psi(\gamma_j, \boldsymbol{\xi}_{(k^{[x]}, k^{[y]})}).$$

- $\gamma_j^{[x]} = e^{-2\pi i x_j}$ ,  $\gamma_j^{[y]} = e^{-2\pi i y_j}$ ,  $\gamma_j = (\gamma_j^{[x]}, \gamma_j^{[y]}) \in \mathbb{S}^2$ ;
- $\xi_{k^{[x]}}^{[x]} = e^{-2\pi i k^{[x]}/n^{[x]}}$ ,  $\xi_{k^{[y]}}^{[y]} = e^{-2\pi i k^{[y]}/n^{[y]}}$ ,  $\boldsymbol{\xi}_{(k^{[x]}, k^{[y]})} = (\xi_{k^{[x]}}^{[x]}, \xi_{k^{[y]}}^{[y]}) \in \mathbb{S}^2$ .

# Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

Submatrix of  $\mathbf{G}$ :

- $\mathcal{R}$ : Row points in  $[a, b) \times [c, d)$ ;
- $\mathcal{C} = \mathcal{C}^{[x]} \times \mathcal{C}^{[y]}$ : Column points in  $[a, b) \times [c, d)$ ;
- $\mathcal{C}^c = \mathcal{K} \setminus \mathcal{C} = \mathcal{C}_b \cup \mathcal{C}_g \cup \mathcal{C}_y$ .



# Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

The submatrix can be partitioned as:

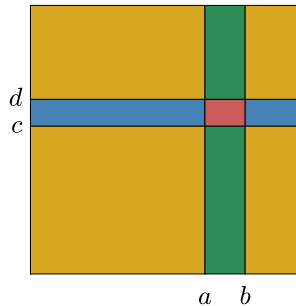
$$\mathbf{G}(\mathcal{R}, \mathcal{C}^c) = [\mathbf{G}(\mathcal{R}, \mathcal{C}_b) \quad \mathbf{G}(\mathcal{R}, \mathcal{C}_g) \quad \mathbf{G}(\mathcal{R}, \mathcal{C}_y)],$$

where

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_b) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x],c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y]})}_{\text{potentially full-rank}},$$

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_g) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x]})}_{\text{potentially full-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y],c})}_{\text{low-rank}},$$

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_y) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x],c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y],c})}_{\text{low-rank}}.$$



# Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

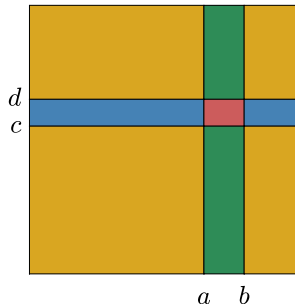
**Lemma:**  $\text{rank}(\mathbf{A}_1 \bullet \mathbf{A}_2) \leq \text{rank}(\mathbf{A}_1) \text{rank}(\mathbf{A}_2)$ .

Therefore, intuitively,

$$\text{rank}_\varepsilon(\mathbf{G}(\mathcal{R}, \mathcal{C}^c)) \lesssim \log(1/\varepsilon) n_c \log n_c,$$

where  $n_c = \max\left\{\left|\mathcal{C}^{[x]}\right|, \left|\mathcal{C}^{[y]}\right|\right\}$ .

$\leadsto \mathbf{G}$  admits an HSS approximation with HSS rank  $\mathcal{O}(\log(1/\varepsilon) \sqrt{N} \log N)$ .



# Face-Splitting Product of HSS Trees

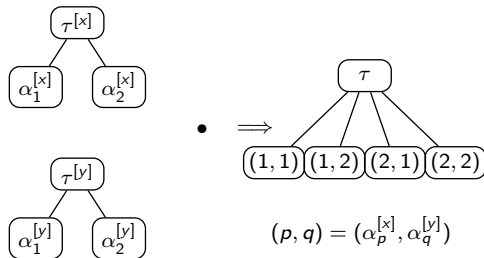
Suppose that  $T^{[x]}$  and  $T^{[y]}$  are full binary HSS trees of the same maximum level.

For  $\tau = (\tau^{[x]}, \tau^{[y]})$ , define

$$\text{ch}(\tau) = \text{ch}(\tau^{[x]}) \times \text{ch}(\tau^{[y]}).$$

The index sets are

$$\mathcal{J}_\tau = \mathcal{J}_{\tau^{[x]}} \cap \mathcal{J}_{\tau^{[y]}} \quad \text{and} \quad \mathcal{K}_\tau = \mathcal{K}_{\tau^{[x]}} \times \mathcal{K}_{\tau^{[y]}}.$$



binary  $\bullet$  binary  $\longrightarrow$  quadtree.

# Closure under the Face-Splitting Product

## Theorem (Li and L., 2026)

*The face-splitting product preserves the HSS structure:*

$\mathbf{T} = \mathbf{T}^{[x]} \bullet \mathbf{T}^{[y]}$  *is an HSS tree,*

$\mathbf{H} = \mathbf{H}^{[x]} \bullet \mathbf{H}^{[y]}$  *is an HSS matrix with respect to  $\mathbf{T}$ .*

## Levelwise rank bound

Suppose that, at level  $\ell$ ,

- the HSS ranks of  $\mathbf{H}^{[x]}$  and  $\mathbf{H}^{[y]}$  are bounded by  $r_\ell^{[x]}$  and  $r_\ell^{[y]}$ ;
- their maximum diagonal-block sizes are  $d_\ell^{[x]}$  and  $d_\ell^{[y]}$ .

Then the HSS rank of  $\mathbf{H}$  at level  $\ell$  is bounded by  $r_\ell^{[x]} d_\ell^{[y]} + r_\ell^{[y]} d_\ell^{[x]} + r_\ell^{[x]} r_\ell^{[y]}$ .

## Theorem (Li and L., 2026)

*Suppose that*

$$\|\tilde{\mathbf{G}}^{[x]} - \mathbf{G}^{[x]}\|_{\mathbf{F}} \leq \varepsilon \|\mathbf{G}^{[x]}\|_{\mathbf{F}} \quad \text{and} \quad \|\tilde{\mathbf{G}}^{[y]} - \mathbf{G}^{[y]}\|_{\mathbf{F}} \leq \varepsilon \|\mathbf{G}^{[y]}\|_{\mathbf{F}}.$$

*Under a mild assumption, the face-splitting product  $\tilde{\mathbf{G}} = \tilde{\mathbf{G}}^{[x]} \bullet \tilde{\mathbf{G}}^{[y]}$  satisfies*

$$\|\tilde{\mathbf{G}} - \mathbf{G}\|_{\mathbf{F}} \leq (1 + C)\varepsilon \|\mathbf{G}\|_{\mathbf{F}}.$$

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# The Construction of the HSS Approximation

**Lemma (Interpolative decomposition, ID, Cheng et al. 2005):** For a rank  $r$  matrix  $\mathbf{A} \in \mathbb{C}^{m \times n}$ , there exists a row subset  $\hat{\mathcal{J}}$  of size  $r$  and an interpolation matrix  $\mathbf{L} \in \mathbb{C}^{m \times r}$  such that  $\mathbf{A} = \mathbf{L}\mathbf{A}(\hat{\mathcal{J}}, :)$ .

To construct the HSS approximation of  $\mathbf{G}$ , we need to compute the row IDs of the HSS row blocks  $\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$  for each node  $\tau$  in the HSS tree:

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) = \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}),$$

where  $\mathcal{Z}_{\mathcal{J}_\tau} = \{\gamma_j\}_{j \in \mathcal{J}_\tau}$  and  $\mathcal{W}_{\mathcal{K}_\tau^c} = \{\xi_{(k^{[x]}, k^{[y]})}\}_{(k^{[x]}, k^{[y]}) \in \mathcal{K}_\tau^c}$ .

However, directly computing the row ID of  $\Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c})$  is expensive because  $|\mathcal{K}_\tau^c|$  is large.

# The Proxy Technique

For each node  $\tau$ , let  $\mathcal{P}_\tau$  be a set of **proxy points** such that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) = \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^*$$

for some matrix  $\mathbf{Q}_\tau$ .

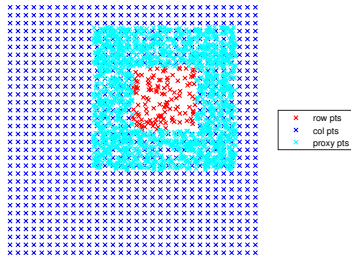
Then, if we compute the row ID

$$\Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{P}_\tau),$$

it follows that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) \approx \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^* \approx \mathbf{L}_\tau \mathbf{G}(\hat{\mathcal{J}}_\tau, \mathcal{K}_\tau^c).$$

Thus, the same interpolation matrix yields a row ID of  $\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$ .



# A Direct Solver for the 2D Type-II Inverse NUDFT Problem

Algebraic structure:

$$\mathbf{A} = \mathbf{GF} \approx \mathbf{A}_{\text{fast}} := \tilde{\mathbf{G}}\mathbf{F},$$

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**Algorithm** A direct solver for the 2D type-II inverse NUDFT problem

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**Input:** Sample points  $\{(x_j, y_j)\}_{j=0}^{M-1}$ , frequency counts  $n^{[x]}$  and  $n^{[y]}$ , target values  $\{f_j\}_{j=0}^{M-1}$ .

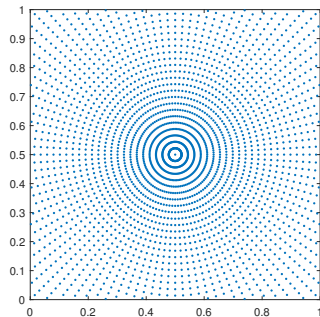
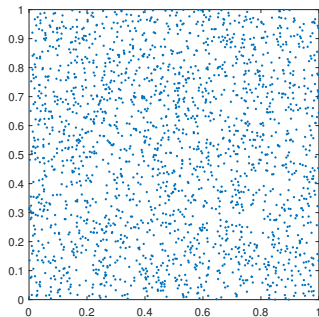
**Output:** Coefficients  $\mathbf{c}$ .

- 1: Construct the HSS approximation  $\tilde{\mathbf{G}}$  of  $\mathbf{G} = \mathbf{AF}^{-1}$  using the kernel matrix expression.
  - 2: Compute the URV factorization of  $\tilde{\mathbf{G}}$ .
  - 3: Compute  $\mathbf{c} = \mathbf{F}^{-1}\tilde{\mathbf{G}}^\dagger \mathbf{f}$  using the URV solve and a two-dimensional iFFT.
- 

Complexities:

- **Offline stage** (construction and factorization):  $\mathcal{O}(M + N^{3/2} \log^3 N)$ ;
- **Online stage** (HSS solve + iFFT):  $\mathcal{O}(M + N \log^3 N)$ .

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- Random grid:  $M \approx 1.5N$ ;
- Polar grid:  $M \approx 0.47N \log_4(N)$ .

# Time Scaling

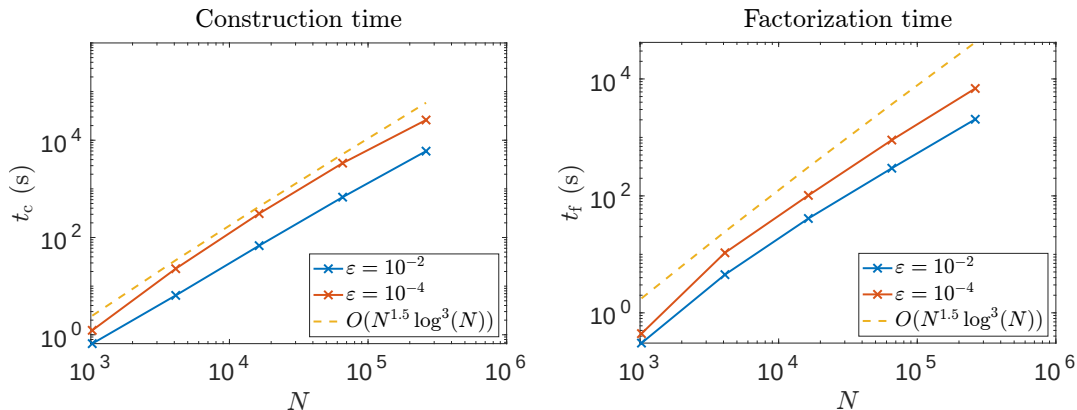


Figure: Construction time and factorization time of the direct solver on a polar grid.

Both times are consistent with the predicted  $\mathcal{O}(N^{3/2} \log^3 N)$  scaling.

# Iterative Solver (Random Grid)

| $N$     | $M$    | method                            | $t_{\text{pre}}$ | $t_{\text{iter}}$ | $n_{\text{iter}}$ | $r_s$   | $e_s$   |
|---------|--------|-----------------------------------|------------------|-------------------|-------------------|---------|---------|
| $64^2$  | 6144   | lsqr                              | -                | 2.7e+00           | 500               | 2.8e-08 | 5.4e-08 |
|         |        | plsqr ( $\varepsilon = 10^{-2}$ ) | 7.2e+00          | 1.0e+00           | 12                | 5.6e-13 | 9.6e-13 |
|         |        | plsqr ( $\varepsilon = 10^{-4}$ ) | 2.0e+01          | 8.9e-01           | 4                 | 1.3e-13 | 2.0e-13 |
| $128^2$ | 24576  | lsqr                              | -                | 4.8e+00           | 500               | 6.5e-05 | 9.4e-05 |
|         |        | plsqr ( $\varepsilon = 10^{-2}$ ) | 8.2e+01          | 8.5e+00           | 14                | 4.3e-12 | 7.0e-12 |
|         |        | plsqr ( $\varepsilon = 10^{-4}$ ) | 3.1e+02          | 7.6e+00           | 6                 | 2.5e-14 | 3.5e-14 |
| $256^2$ | 98304  | lsqr                              | -                | 1.2e+01           | 500               | 1.2e-04 | 2.1e-04 |
|         |        | plsqr ( $\varepsilon = 10^{-2}$ ) | 7.7e+02          | 6.6e+01           | 23                | 2.0e-12 | 3.1e-12 |
|         |        | plsqr ( $\varepsilon = 10^{-4}$ ) | 3.4e+03          | 5.6e+01           | 8                 | 8.8e-13 | 1.3e-12 |
| $512^2$ | 393216 | lsqr                              | -                | 3.0e+01           | 500               | 1.9e-04 | 3.2e-04 |
|         |        | plsqr ( $\varepsilon = 10^{-2}$ ) | 6.5e+03          | 5.1e+02           | 39                | 3.6e-12 | 5.5e-12 |
|         |        | plsqr ( $\varepsilon = 10^{-4}$ ) | 2.9e+04          | 4.6e+02           | 14                | 2.7e-12 | 4.0e-12 |

lsqr reaches the 500-iteration limit, whereas plsqr converges in **fewer than 40 iterations**.

# Iterative Solver (Polar Grid)

| $N$     | $M$     | method                            | $t_{\text{pre}}$ | $t_{\text{iter}}$ | $n_{\text{iter}}$ | $r_s$   | $e_s$   |
|---------|---------|-----------------------------------|------------------|-------------------|-------------------|---------|---------|
| $64^2$  | 11620   | lsqr                              | -                | 2.9e+00           | 500               | 3.8e-03 | 4.0e-03 |
|         |         | plsqr ( $\varepsilon = 10^{-2}$ ) | 1.1e+01          | 2.2e+00           | 21                | 1.5e-11 | 1.6e-11 |
|         |         | plsqr ( $\varepsilon = 10^{-4}$ ) | 3.3e+01          | 1.4e+00           | 5                 | 8.9e-11 | 9.3e-11 |
| $128^2$ | 54373   | lsqr                              | -                | 5.6e+00           | 500               | 2.2e-05 | 1.6e-03 |
|         |         | plsqr ( $\varepsilon = 10^{-2}$ ) | 1.1e+02          | 2.8e+01           | 38                | 2.6e-13 | 1.4e-11 |
|         |         | plsqr ( $\varepsilon = 10^{-4}$ ) | 4.1e+02          | 1.5e+01           | 10                | 2.2e-13 | 6.7e-11 |
| $256^2$ | 249074  | lsqr                              | -                | 1.7e+01           | 500               | 1.9e-03 | 1.8e-03 |
|         |         | plsqr ( $\varepsilon = 10^{-2}$ ) | 9.7e+02          | 1.3e+02           | 37                | 1.3e-11 | 1.2e-11 |
|         |         | plsqr ( $\varepsilon = 10^{-4}$ ) | 4.3e+03          | 1.0e+02           | 12                | 5.7e-11 | 5.5e-11 |
| $512^2$ | 1122019 | lsqr                              | -                | 6.5e+01           | 500               | 4.8e-04 | 4.6e-04 |
|         |         | plsqr ( $\varepsilon = 10^{-2}$ ) | 8.0e+03          | 4.2e+02           | 24                | 2.2e-12 | 2.1e-12 |
|         |         | plsqr ( $\varepsilon = 10^{-4}$ ) | 3.3e+04          | 4.2e+02           | 10                | 2.5e-12 | 2.4e-12 |

lsqr reaches the 500-iteration limit, whereas plsqr converges in **fewer than 40 iterations**.

- **Decomposition:**

$$\mathbf{A} = \mathbf{GF} \approx \tilde{\mathbf{G}}\mathbf{F}$$

$\mathbf{G}$ : kernel matrix;  $\mathbf{F}$ : 2D DFT matrix;  $\tilde{\mathbf{G}}$ : HSS approximation.

- **Direct solver:**

Offline complexity  $\mathcal{O}(M + N^{3/2} \log^3 N)$  and online complexity  $\mathcal{O}(M + N \log^3 N)$ ;

- **Effective preconditioner:**

On polar grids, the iteration count drops from the maximum of 500 to **fewer than 40**.

# Thank you!

My homepage:

<https://jingyuliumath.github.io/>

Related paper:

<https://arxiv.org/abs/2607.00928>

